

PROJECT 2-STAGE OTA

Two-Stage OTA Design in SKY130A

Specification-Driven Analog Design from Ideal Architecture to Transistor Level

PDK	SKY130A
Tools	Xschem · ngspice · IIC-OSIC-TOOLS
Revision	0.1
Date	September 2026
Repository	GitHub repository

Guillaume Monti

Analog IC Design Engineer

Contents

1	Project Overview	1
2	Specifications and Architecture	2
2.1	Initial Design Specifications	2
2.2	Selected Topology	2
2.3	Device Naming	2
2.4	Bias-Current Architecture	3
3	First-Order Design	4
3.1	Bias-Current Selection	4
3.2	Input-Stage Transconductance	4
3.3	Miller Compensation Capacitor	4
3.4	Second-Stage Transconductance	5
3.5	Second-Stage Bias Ratio	5
3.6	First-Order Slew-Rate Estimate	6
3.7	Initial Transistor Sizing	6
3.8	First-Order Gain Estimate	8
4	Ideal Two-Stage Model	10
4.1	Nominal AC Response	10
4.2	Load-Capacitance Sensitivity	11
4.3	Miller Zero Compensation	11
5	Transistor-Level Design	13
6	Characterization	14
7	Conclusion	15

1 Project Overview

This project develops a classical two-stage CMOS OTA using the SKY130A open-source PDK.

The intended application is a low-power analog block representative of PMIC and voltage-reference circuits, where DC gain, low current consumption, stability and supply rejection are prioritized over maximum speed.

The design follows a specification-driven methodology. First-order analytical calculations are used to establish a coherent initial design before transistor sizing. An ideal architectural model is then used to study the fundamental two-stage behavior and compensation strategy before implementing the complete transistor-level circuit.

Transistor sizing reuses the SKY130A device characterization obtained in the preceding `project_carac_sky130` project, including g_m/I_D , I_D/W , intrinsic gain, threshold-voltage and saturation-voltage data.

The design flow used throughout the project is:

1. define the OTA specifications,
2. select the topology and bias-current architecture,
3. derive a first-order analytical design,
4. study the ideal two-stage architecture and compensation,
5. size the SKY130A transistors using the characterized device data,
6. verify and iterate at transistor level,
7. characterize the final OTA over the relevant operating conditions.

2 Specifications and Architecture

2.1 Initial Design Specifications

The OTA is intended for a low-power PMIC or voltage-reference application. The design therefore prioritizes DC gain, low current consumption, stability and supply rejection over maximum speed.

The initial design targets are summarized in Table 1.

Table 1 Initial OTA design specifications.

Parameter	Initial target
Supply voltage	1.8 V
DC gain	≥ 80 dB
GBW	≥ 1 MHz
Phase margin	$\geq 60^\circ$
Load capacitance	2 pF nominal, 4 pF maximum
Quiescent current	Minimize within the other constraints
PSRR	High priority, characterized versus frequency
Slew rate	Secondary design objective

2.2 Selected Topology

A classical two-stage Miller-compensated CMOS OTA is selected.

The first stage uses a PMOS differential pair with an NMOS current-mirror active load. Its single-ended output node, `out_s1`, drives the second stage.

The second stage consists of the NMOS common-source transistor `NLOAD_2`, biased by the PMOS current source `PTAIL_2`. The OTA output is denoted `Vout`.

The compensation network is connected between `out_s1` and `Vout`. It contains the Miller capacitor `CMILLER` and, when required, the series resistor `RMILLER`.

2.3 Device Naming

The naming convention shown in Figure 1 is used throughout the analytical design and simulation results.

- `PTAIL_0`: diode-connected PMOS receiving the reference bias current and generating the PMOS mirror bias voltage,
- `PTAIL_1`: PMOS current source biasing the first stage,
- `PDIFF_P`, `PDIFF_N`: PMOS input differential pair,
- `NLOAD_0`, `NLOAD_1`: NMOS current-mirror active load,
- `out_s1`: first-stage output and second-stage input,
- `PTAIL_2`: PMOS current source biasing the second stage,
- `NLOAD_2`: NMOS common-source transistor of the second gain stage,
- `Vout`: OTA output,

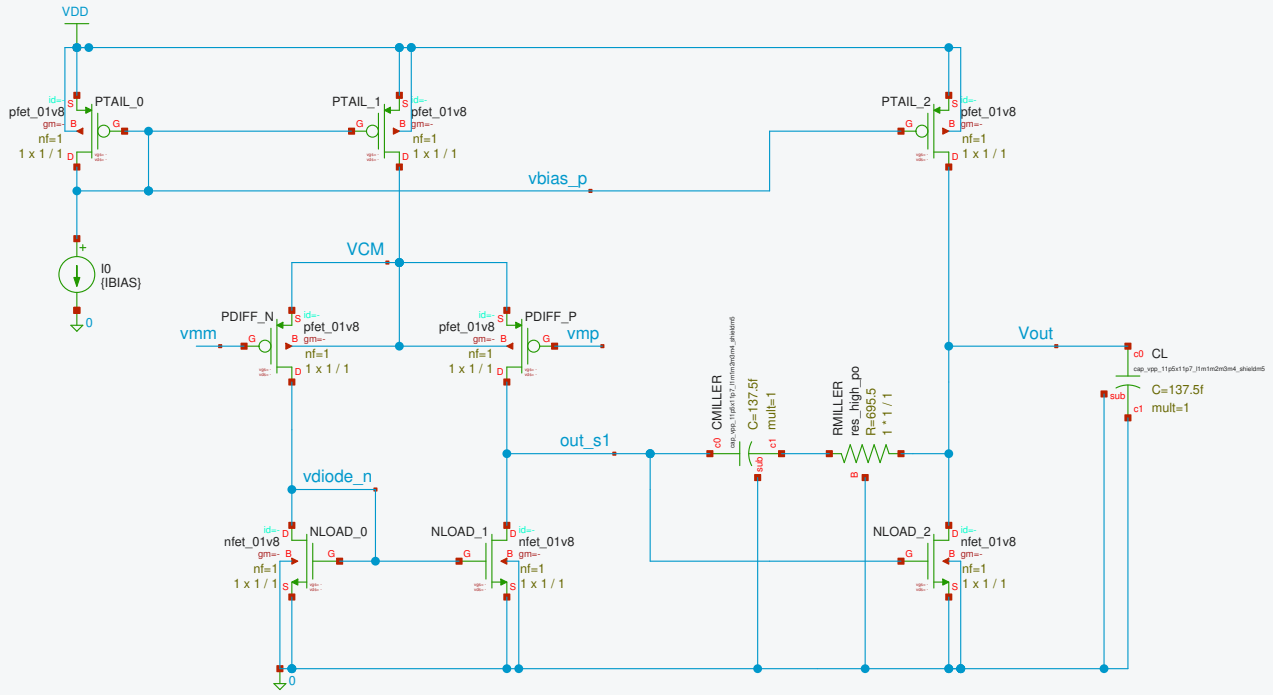


Figure 1 Selected two-stage OTA topology and device naming convention.

- CMILLER: Miller compensation capacitor,
- RMILLER: optional series compensation resistor.

2.4 Bias-Current Architecture

The diode-connected device PTAIL_0 defines the reference current unit I .

The first-stage current is initially defined as

$$I_{PTAIL_1} = I,$$

which gives, at zero differential input,

$$I_{PDIFF_P} = I_{PDIFF_N} = \frac{I}{2}.$$

The second-stage current is expressed as

$$I_{PTAIL_2} = NI,$$

where the mirror ratio N is derived from the required second-stage transconductance and stability constraints in the first-order design.

3 First-Order Design

The OTA is first sized analytically to obtain a coherent initial operating point before transistor-level optimization. These calculations are intended as first-order design guidance rather than exact predictions of the final BSIM-based circuit.

3.1 Bias-Current Selection

The diode-connected PMOS P_{TAIL_0} defines the reference bias current

$$I = 500 \text{ nA}.$$

The first-stage PMOS current source is initially mirrored with a ratio of one:

$$I_{PTAIL_1} = I = 500 \text{ nA}.$$

At zero differential input, the differential-pair current therefore splits equally:

$$I_{PDIFF_P} = I_{PDIFF_N} = \frac{I}{2} = 250 \text{ nA}.$$

The second-stage bias current is expressed as a multiple of the reference current:

$$I_{PTAIL_2} = NI.$$

The value of N is derived from the required second-stage transconductance and stability constraints.

3.2 Input-Stage Transconductance

For the low-power input differential pair, an initial transconductance efficiency of

$$\frac{g_m}{I_D} \approx 20 \text{ V}^{-1}$$

is selected.

With

$$I_D = 250 \text{ nA},$$

the expected input-device transconductance is

$$g_{m1} = \frac{g_m}{I_D} I_D = 20 \times 250 \text{ nA} = 5 \text{ } \mu\text{S}.$$

3.3 Miller Compensation Capacitor

For a classical Miller-compensated two-stage OTA, the unity-gain bandwidth is approximated by

$$GBW \approx \frac{g_{m1}}{2\pi C_{MILLER}}.$$

Using the target

$$GBW = 1 \text{ MHz},$$

the required compensation capacitance is

$$C_{\text{MILLER}} = \frac{5 \mu\text{S}}{2\pi \times 1 \text{ MHz}} \approx 0.80 \text{ pF}.$$

The initial compensation value is therefore

$$C_{\text{MILLER}} \approx 0.8 \text{ pF}.$$

A SKY130A MIM capacitor is used for the physical implementation of this component.

3.4 Second-Stage Transconductance

The second-stage transconductance must place the non-dominant output pole sufficiently above the unity-gain frequency.

Using the first-order approximation

$$\omega_{p2} \approx \frac{g_{m2}}{C_L}$$

and targeting

$$\omega_{p2} \approx 2.2 \omega_u$$

for approximately 60° phase margin gives

$$g_{m2} \approx 2.2 g_{m1} \frac{C_L}{C_{\text{MILLER}}}.$$

For the worst-case initial load

$$C_L = 4 \text{ pF},$$

the required second-stage transconductance is approximately

$$g_{m2} \approx 2.2 \times 5 \mu\text{S} \times \frac{4 \text{ pF}}{0.8 \text{ pF}} = 55 \mu\text{S}.$$

The initial target is therefore

$$g_{m2} \approx 55\text{--}60 \mu\text{S}.$$

3.5 Second-Stage Bias Ratio

An initial second-stage current of

$$4I = 2 \mu\text{A}$$

would require

$$\frac{g_m}{I_D} \approx \frac{55 \mu\text{S}}{2 \mu\text{A}} = 27.5 \text{ V}^{-1}.$$

This operating point is unnecessarily aggressive for the initial design.

Increasing the mirror ratio to

$$N = 6$$

gives

$$I_{\text{PTAIL}_2} = 6I = 3 \mu\text{A}.$$

The corresponding required transconductance efficiency becomes

$$\frac{g_m}{I_D} \approx \frac{55 \mu\text{S}}{3 \mu\text{A}} \approx 18.3 \text{ V}^{-1},$$

which is a more suitable initial operating region.

The initial bias-current ratio is therefore selected as

$$I_{\text{PTAIL}_0} : I_{\text{PTAIL}_1} : I_{\text{PTAIL}_2} = 1 : 1 : 6.$$

3.6 First-Order Slew-Rate Estimate

Although slew rate is not a primary design objective, it provides a useful large-signal sanity check.

Assuming that the first-stage bias current limits the charging or discharging of the Miller capacitor,

$$SR \approx \frac{I_{\text{PTAIL}_1}}{C_{\text{MILLER}}}.$$

Therefore,

$$SR \approx \frac{500 \text{ nA}}{0.8 \text{ pF}} \approx 0.63 \text{ V}/\text{s}.$$

This value is considered sufficient as an initial estimate for the intended low-power precision application.

3.7 Initial Transistor Sizing

The transistor dimensions are obtained from the SKY130A device characterization performed in the preceding project.

For each device group, the drain current, target g_m/I_D and channel length are first selected from the circuit-level requirements. The characterized current density I_D/W is then used to determine the required width.

3.7.1 Input Differential Pair

The PMOS differential pair carries

$$I_D = 250 \text{ nA}$$

per device and targets

$$\frac{g_m}{I_D} \approx 20 \text{ V}^{-1}.$$

A channel-length comparison showed that increasing the length from $1 \mu\text{m}$ to $2 \mu\text{m}$ strongly increased device area for comparatively limited circuit-level benefit.

The initial geometry is therefore selected as

$$W_{\text{PDIFF}} = 89 \mu\text{m}, \quad L_{\text{PDIFF}} = 1 \mu\text{m}.$$

At this operating point, the characterized intrinsic gain is approximately

$$\frac{g_m}{g_{ds}} \approx 478.$$

3.7.2 First-Stage NMOS Mirror Load

The two NMOS mirror devices also carry

$$I_D = 250 \text{ nA}.$$

A preliminary target of $g_m / I_D \approx 15 \text{ V}^{-1}$ resulted in a calculated width below the minimum practical device width.

The width was therefore fixed to

$$W = 0.42 \mu\text{m},$$

and the corresponding operating point was re-evaluated from the characterized data.

For $L = 1 \mu\text{m}$, the resulting values are approximately

$$\frac{g_m}{I_D} \approx 19.5 \text{ V}^{-1}$$

and

$$\frac{g_m}{g_{ds}} \approx 210.$$

The initial geometry is therefore

$$W_{\text{NLOAD0,1}} = 0.42 \mu\text{m}, \quad L_{\text{NLOAD0,1}} = 1 \mu\text{m}.$$

3.7.3 Second-Stage Gain Device

The NMOS second-stage gain transistor carries

$$I_D = 3 \mu\text{A}$$

and targets

$$\frac{g_m}{I_D} \approx 18.3 \text{ V}^{-1}.$$

The characterized data showed little intrinsic-gain improvement between $L = 1 \mu\text{m}$ and $L = 2 \mu\text{m}$, while the required width approximately doubled.

The selected initial geometry is therefore

$$W_{\text{NLOAD2}} = 3.58 \mu\text{m}, \quad L_{\text{NLOAD2}} = 1 \mu\text{m}.$$

3.7.4 PMOS Bias Mirror

The PMOS bias devices are operated around

$$\frac{g_m}{I_D} \approx 15 \text{ V}^{-1}.$$

For the reference current of 500 nA, the selected geometry is

$$W_{\text{PTAIL0}} = W_{\text{PTAIL1}} = 1.76 \mu\text{m}, \quad L = 1 \mu\text{m}.$$

The second-stage current source uses the same operating point and a 6 : 1 mirror ratio:

$$W_{\text{PTAIL2}} = 6 \times 1.76 = 10.56 \mu\text{m}.$$

3.7.5 Initial Sizing Summary

Table 2 Initial transistor sizing obtained from first-order design and SKY130A characterized device data.

Device	I_D	g_m/I_D	$W (\mu\text{m})$	$L (\mu\text{m})$
PTAIL_0	500 nA	14.9	1.76	1.0
PTAIL_1	500 nA	14.9	1.76	1.0
PDIFF_N/P	250 nA	20.0	89.0	1.0
NLOAD_0/1	250 nA	19.5	0.42	1.0
PTAIL_2	3 μA	14.9	10.56	1.0
NLOAD_2	3 μA	18.2	3.58	1.0

3.8 First-Order Gain Estimate

Using the characterized output conductances at the selected operating points, the first-stage gain is estimated as

$$A_{v1} \approx g_{m1} (r_{o,\text{PDIFF}} \parallel r_{o,\text{NLOAD1}}) \approx 43.5 \text{ dB}.$$

For the second stage,

$$A_{v2} \approx g_{m2} (r_{o,\text{NLOAD2}} \parallel r_{o,\text{PTAIL2}}) \approx 43.7 \text{ dB}.$$

The expected total low-frequency gain is therefore

$$A_0 \approx A_{v1} + A_{v2} \approx 87 \text{ dB}.$$

This provides approximately 7 dB of nominal first-order margin relative to the 80 dB design target. The result will subsequently be verified using transistor-level simulation.

4 Ideal Two-Stage Model

An ideal small-signal model is first used to verify the analytical two-stage design independently of MOS transistor parasitics.

Each gain stage is represented by a voltage-controlled current source and its corresponding output resistance. The model parameters are directly obtained from the first-order transistor sizing:

$$g_{m1} = 5 \mu\text{S}, \quad R_{\text{out}1} = 29.7 \text{ M}\Omega$$

and

$$g_{m2} = 54.4 \mu\text{S}, \quad R_{\text{out}2} = 2.81 \text{ M}\Omega.$$

The two stages are compensated using

$$C_{\text{MILLER}} = 0.8 \text{ pF}.$$

No additional internal capacitance is introduced at this stage. The objective is to isolate the behavior predicted by the first-order two-stage model. The external load capacitance is applied from the testbench.

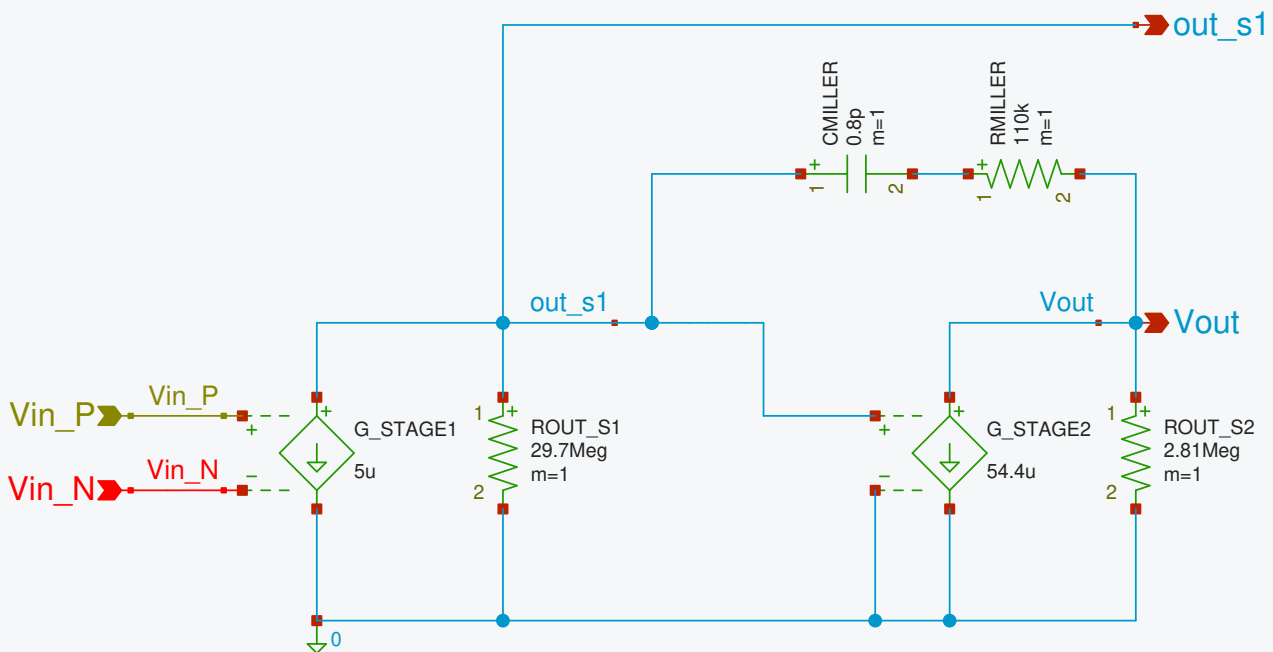


Figure 2 Ideal two-stage OTA model using transconductance stages, calculated output resistances and Miller compensation.

4.1 Nominal AC Response

The initial AC simulation uses

$$C_L = 2 \text{ pF}.$$

The simulated low-frequency gain is

$$A_0 = 87.1 \text{ dB},$$

while the unity-gain frequency is

$$f_{\text{UGF}} = 966.7 \text{ kHz}.$$

The phase at unity gain is

$$\phi(f_{\text{UGF}}) = -107.6^\circ,$$

corresponding to a phase margin of

$$PM = 72.4^\circ.$$

The ideal model therefore reproduces the analytical gain and bandwidth predictions closely while providing comfortable margin relative to the 60° stability target.

4.2 Load-Capacitance Sensitivity

The load capacitance was increased from the nominal value of 2 pF to the specified maximum value of 4 pF, without modifying the OTA parameters.

The resulting AC performance is summarized in Table 3.

Table 3 Ideal-model AC performance versus load capacitance.

C_L	DC gain	UGF	Phase margin
2 pF	87.1 dB	966.7 kHz	72.4°
4 pF	87.1 dB	912.0 kHz	62.5°

As expected, the low-frequency gain is essentially unaffected by the capacitive load. The increased load capacitance lowers the non-dominant pole and therefore reduces the phase margin.

At $C_L = 4 \text{ pF}$, the model retains approximately 62.5° of phase margin and therefore still satisfies the initial 60° stability target, although with limited additional margin.

4.3 Miller Zero Compensation

Without a series compensation resistor, the Miller feedforward path introduces a right-half-plane zero approximately located at

$$f_{z,\text{RHP}} \approx \frac{g_{m2}}{2\pi C_{\text{MILLER}}} \approx 10.8 \text{ MHz}.$$

For a series Miller resistor, the first-order zero location depends on

$$R_{\text{MILLER}} = \frac{1}{g_{m2}}.$$

For

$$R_{\text{MILLER}} \approx \frac{1}{g_{m2}} \approx 18.4 \text{ k}\Omega,$$

the RHP zero is approximately pushed to infinity. At the maximum 4 pF load, this increases the phase margin from 62.5° to 67.4° with negligible bandwidth penalty.

A larger resistance creates a left-half-plane zero. To compensate the phase lag introduced by the non-dominant output pole, the zero was placed approximately around

$$f_{p2} \approx \frac{g_{m2}}{2\pi C_L} \approx 2.2 \text{ MHz}.$$

This gives an initial resistor estimate of approximately

$$R_{\text{MILLER}} \approx 110 \text{ k}\Omega.$$

The resulting LHP zero provides phase lead around the second-pole frequency, making the open-loop response behave approximately as a single-pole system around unity gain without increasing C_{MILLER} .

Table 4 Ideal-model compensation and load study.

C_L	R_{MILLER}	DC gain	UGF	PM
2 pF	0 Ω	87.1 dB	966.7 kHz	72.4°
4 pF	0 Ω	87.1 dB	912.0 kHz	62.5°
4 pF	18.4 k Ω	87.1 dB	909.1 kHz	67.4°
4 pF	110 k Ω	87.1 dB	985.2 kHz	90.1°

The 110 k Ω case demonstrates the intended pole-zero compensation mechanism particularly clearly: the LHP zero approximately counteracts the phase lag of the second pole around the relevant frequency range.

This value is therefore retained as the initial compensation-resistor estimate for the transistor-level design. Its optimum value is expected to shift once transistor parasitics and additional high-frequency poles are introduced.

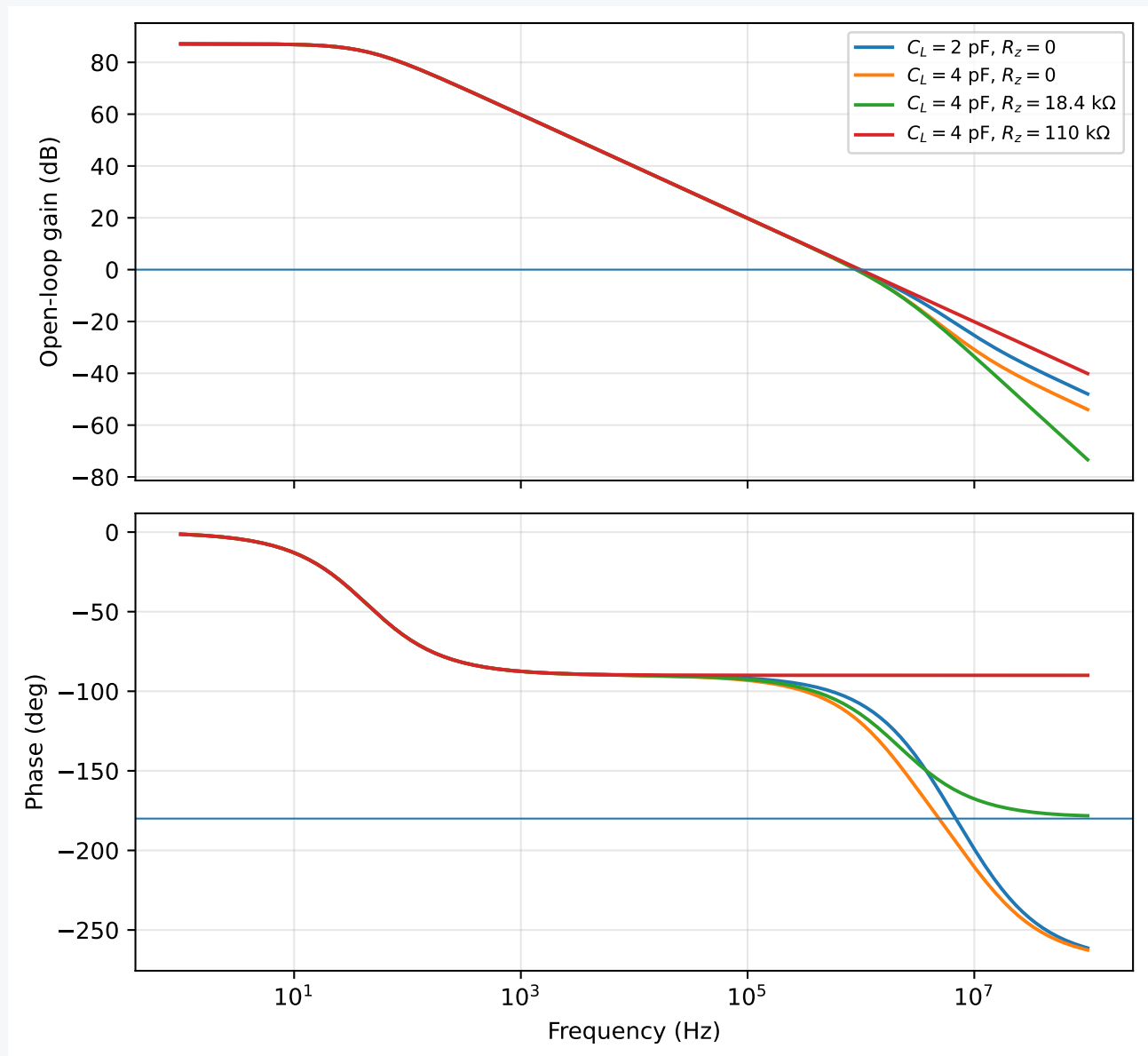


Figure 3 Ideal open-loop response showing the effect of load capacitance and Miller zero compensation.

5 Transistor-Level Design

6 Characterization

7 Conclusion